Project Intro

Lec 12

Last Time

- Discussed how AI interventions are implemented in the clinic
- Thought through different high-level implementations of ideas
  - Problem type
  - Training regime
  - Input/output data
- FDA device approval

Big Idea

- Come up with an AI intervention of your own creation
- Based on a clinical question, ideally from your specialty
- Should be novel and not something already done

Part 1: Problem Formulation

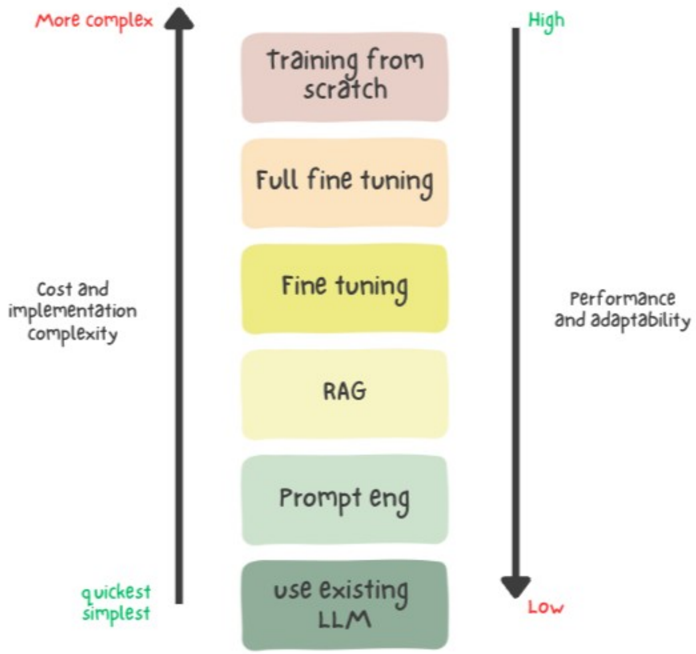
- Think about various workflows in your field.
  - How are diagnoses made?
  - Given a diagnosis, how do we decide which treatment is appropriate?
  - How are patient encounters/histories/diagnoses/treatments documented?
  - Are there logistic challenges that could be addressed?
- Develop a clinical question that addresses a challenge in your field
- Ensure this question is novel
  - Search arxiv and pubmed to ensure this idea hasn't been tried before
  - Note: if implementation is novel enough you can repeat idea

Part 1: Problem Formulation

- Deliverables:
  - What is the clinical question?
  - One paragraph on the relevance of this question to your field
  - Few sentences on related literature and what has thus far been tried, either in related conditions or with similar approaches and why your approach is novel
- Note: Specific is better, should include input data and outputs
- Examples:
  - Can we predict sepsis at any point in emergency room patients based on clinical data collected in the first 24 hours?
  - Based on vital signs at an outpatient visit, can we predict a patient’s hemoglobin level?
  - Using patient interview transcripts, can we predict their MOCA

Part 2: Implementation

- What AI approach will be used to address this question?
- Task:
  - Classification (multiclass) (multi single class)
  - Regression
  - Does your task have a special simplified name? i.e. image segmentation = pixel-based classification
  - Generative?
  - Don't confuse LLM training with prompting



Part 2: Implementation

- Deliverable: What is the technical problem statement?
  - Deep learning based classification of patients into sepsis and sepsis free groups based on data from first 24 h of admission
  - Prediction of patient hemoglobin levels using deep learning regression based on vital signs at outpatient visit
  - Segmentation of soft tissue sarcoma using multimodal CT-MRI-PET data
  - Prediction of patient MOCA scores using deep learning based regression on language embedding encoded interview transcripts
- Deliverable: Few sentences on why this approach to implementation makes sense.

Part 3: Data Sourcing

- What data will you need and why? Ex. If you are doing prediction of hemoglobin, do you need their IgM level?
- How will you acquire the data used in this project?
- Are there any public datasets that can be used? If not, why?
- Will the data need to be cleaned, standardized or preprocessed at all? Will we remove outliers? Should some part of the input data be discarded? What biases might your approach of data gathering lead to?
- Deliverable: Answer the above questions in a paragraph

Part 4: Additional Technical Dive

- Deliverable: A paragraph on **one** of these technical areas and what the considerations will be in your intervention

Data	Loss
What is the sample population? Type of data - image, chart, time series, multimodal How is data encoded? Numeric, thermometer, 1hot or other? Augmentations? Ground truth? Experts, lay people, other algorithm	Classically, should match problem type: - Regression: MSE, Classification: Log loss/Multiclass log loss - Why does this make sense? What is loss function rewarding? Consider regulari-zation.
Network	**Training**
Classic neural net or with variations? autoencoder, GAN, RNN, LSTM, CNN, combo of these. If not sure, what's it closest to? If brand new, what kind of data can it take in and process well?	What hyperparameters were considered? Should have to do with the network structure Use of pretrained model weights? Train/test split—random, cross validation

Part 5: Deployment and clinical relevance

- Deliverable: Few sentences on output metrics that will be tracked to determine and monitor the success of the algorithm
  - F1, AUROC, Sensitivity, Specificity, PPV, ballpark numbers that would be acceptable
- Deliverable: Paragraph on potential pitfalls to rollout in the clinic. What would biased results look like in the final model? Could the results slow down workflows in an unexpected way? How will periodic surveillance for data drift be implemented?

Use of AI

- Let AI be your assistant, not your replacement
- Use AI as you see fit, but be ready to explain your choices verbally
- Any work AI does that you turn in you place your seal of approval on, including errors
- Last day of class will call on some people randomly to discuss various decisions they’ve made

Grading

- 10/10 possible on each section, maximum 50/50
- Will be graded on correctness, clinical relevance and mastery of technical knowledge
- Thoughtful responses that show clear effort will definitely pass
- 70% is passing, 50% of overall course grade
- Will be averaged with attendance percentage to determine final grade
- Bonus up to 10% if you give a 10-15 min powerpoint presentation to the class on last day
- Reserve the right to award lower scores on a given section if it is blatantly AI generated with zero edits + cannot explain when asked

Final Thoughts

- You get out what you put in
- Work hard on this and it could become a real research question you attempt to solve!
- Matthew and myself are available after class for up to 1 hour to answer any questions that come up